Design of an AR-HMD Interface for Interactive Control of Robotic Tack and Seam Welding

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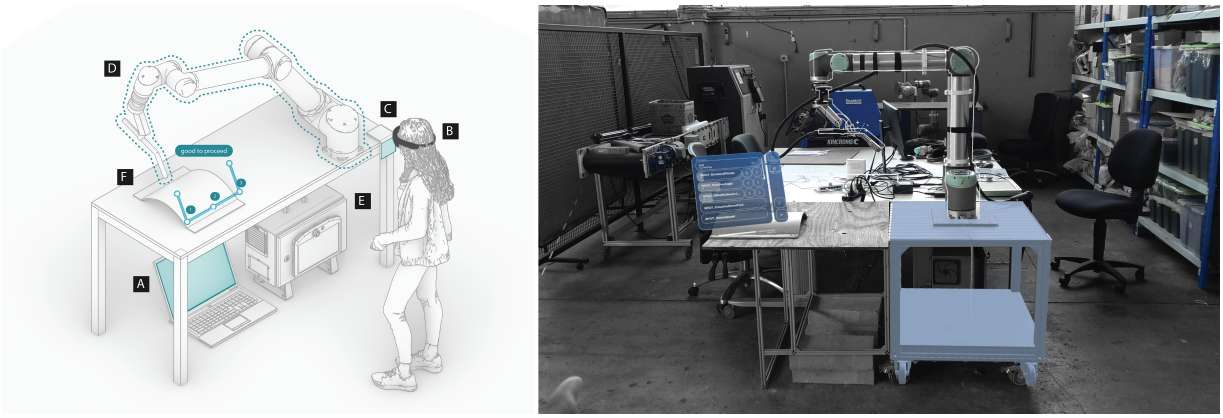


Figure 1: [Left] Proposed AR-assisted robotic welding setup. (A) Workstation, (B) Microsoft HoloLens 2, (C) Visual marker, (D) UR10, collaborative robot, and (E) Control box. [Right] Tack welding mode within the proposed AR interface

Abstract

Current augmented reality (AR) approaches for robotic welding programming often rely on programming through demonstration (PtD), which can be time-consuming and inflexible—particularly in complex, variable manufacturing environments. Furthermore, prior research has primarily addressed seam welding, with little focus on tack welding or workflows that integrate both. We present a proof-of-concept AR system that enables interactive planning of seam and tack welding tasks through gesture-based spatial input. Users define weld points directly on physical workpieces within an AR environment, with real-time overlays visualising the robot's reachability and path constraints. This approach facilitates in-situ validation and supports a human-in-the-loop (HILT) planning process. An informal evaluation conducted with members of the research team. The preliminary system level validation demonstrates feasibility and suggests potential for future HITL evaluation. This paper presents a novel approach that make two contribution: (1) a unified AR-based computational workflow for interactive seam and tack welding programming, and (2) a prototype interface leveraging real-time spatial feedback and natural gestural input for robotic welding.

Index Terms: Augmented Reality (AR), Head-Mounted Display (HMD), Human-Robot Interaction (HRI), Robotic Welding, System Design.

1 Introduction and Related Work

Welding is an essential, yet physically demanding and hazardous profession, requiring significant precision and expertise [3]. Despite its critical role in manufacturing and construction settings, industries face a steady decline in skilled welders, driven largely by occupational health risks and physical strain [1]. In recent years, the robotic welding system was proposed as a viable solution to improve workers' safety, productivity, and consistency. However, their adoption remains limited due to the complexity and specialised robotics expertise required, which most skilled welders lack. Furthermore, conventional robotic programming for welding tasks often involves using teach pendant or CAD-based programming, which are time consuming and require accurate representation of the virtual environment [6].

Augmented reality (AR) has recently emerged as a promising solution to overcome these limitations, as it simplifies robotic welding programming through intuitive, spatially contextualised interfaces [5]. The AR was deployed in tandem with Programming-through-Demonstration (PtD) for robotic seam welding operation. As demonstrated in [5], the authors use this approach to support non-expert users to define welding paths. It is important to note that the accuracy of PtD heavily depends on the quality of the demonstration and the capability of the motion capture tool. Recognising this limitation, [4] have explored this interaction with path optimisation strategies to enhance the precision of the demonstrations.

Although most existing research on AR-supported welding is often associated with PtD , recent advances in computational programming allow us to diverge from this paradigm. By leveraging these computational capabilities, we propose an AR-based method that directly defines robotic welding paths. In parallel, we observe that the literature predominantly focuses on seam welding, while tack welding remains relatively unexplored despite its significant role in the assembly process. Our work addresses this gap by applying our computationally driven AR approach specifically to tack welding.

In this work, we present a proof-of-concept AR-based system that allows users to intuitively define both tack and seam welding paths without robotics expertise. To the best of our knowledge, this study is the first to specifically address AR-assisted tack welding, thereby expanding the scope of AR-supported human-robot interaction (HRI) research into a crucial yet underexplored area.

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2 System Design

2.1 System Architecture: Hardware

The hardware system comprises three main components: an AR-HMD, a cobot, and a Window PC (illustrated in Figure 1). Microsoft HoloLens 2 was used to facilitate immersive visualisation, and hands-free, spatial interaction with weld paths through the Fologram [2] application. The robotic workcell includes a collaborative robot (cobot), the UR10 CB-series, which executes the welding path defined by the users.

The workstation and cobot are connected via an Ethernet cable to ensure reliable transmission of robot commands. At the same time, the AR-HMD communicates with the workstation over a stable local Wi-Fi network. The AR visual environment is spatially anchored on the cobot's physical workspace via a visual marker.

2.2 System Architecture: Software

The system is built on *Grasshopper*, a visual programming platform embedded in Rhino 3D. Weld points and seam paths are generated using geometry-based definitions, and robot trajectories are planned through the Robots plugin, which supports the simulation and export of URScript-compatible code.

Fologram [2] bridges the AR and programming environments by streaming the Grasshopper model to the *HoloLens 2* in real time, allowing users to view and interact with weld plans overlaid on the physical workpiece. All parameter changes in the AR environment are synchronised with the underlying Grasshopper model.

2.3 Interaction Workflow

Users interact with the system through an AR interface rendered on the *Microsoft HoloLens 2*. An on-screen menu allows users to switch between tack and seam welding modes. Within each mode, users can adjust design parameters such as (A) welding speed, (B) welding angle, (C) offset from the metal workpieces, and (D) resolution (number of target points). In addition, users can define the number of tack welds while in the tack welding modes.

Weld paths are defined by manipulating the virtual widgets using pinch-and-drag gestures. These widgets are directly associated with robot poses and tool orientations. As users adjust cobot waypoint(s), a real-time trajectory preview is rendered in the visual environment, providing immediate visual feedback on the planned toolpath.

The AR environment also incorporates colour-coded visual cues and text annotation (refer to Figure 2) to indicate reachability constraints, helping users detect and correct infeasible paths before execution. Once validated, users transmit the command directly from the AR interface to the cobot, completing the planning execution loop.

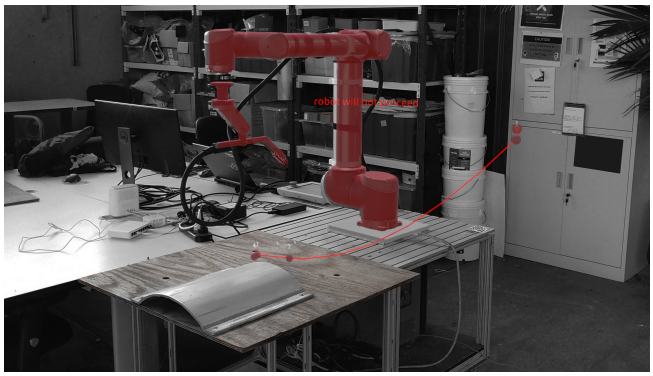


Figure 2: Color-coded visual cues and text annotation appeared within user's FOV to indicate reachability constraints.

3 Results and Discussion

An informal test of the proof-of-concept was carried out within the research team to explore the feasibility of the system. These early findings suggest the system's potential for streamlining robotic welding setup, particularly in environments where flexibility and rapid prototyping are critical. One challenge observed was maintaining consistent spatial anchoring between the AR environment and the physical workspace, especially during long sessions.

Although geometry-based interactions effectively supported tack welding, seam welding may benefit more from traditional AR-supported PtD as shown in [4, 5]. Given that welding practice involves significant tacit knowledge, including embodied skills and spatial intuition, PtD could capture nuances that our interaction approach may overlook. Nonetheless, features such as real-time reachability feedback remain valuable for users with limited robotics expertise.

4 Conclusion

This poster presented a proof-of-concept system that integrates an AR-based interaction for tack and seam welding tasks. By allowing users to define weld paths directly within a spatially anchored AR environment using gestural inputs, the system offers a more simplified and flexible approach to robot programming.

Future work could adopt a more user-centered design approach by involving skilled welders in iterative system development to ensure that the developed system meets their practical needs. We also plan to conduct formal studies to evaluate task efficiency, learning curves, and cognitive load.

Supplemental Materials

Supplementary materials are available on Github at https://github.com/LoyWeiWin/Pub_ISMAR_RoboticWelding, released under a GPL V3 license. In particular, they include (1) figure images in full resolution, (2) low resolution of the videos, (3) a full version of this article, and (4) the system developed.

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